



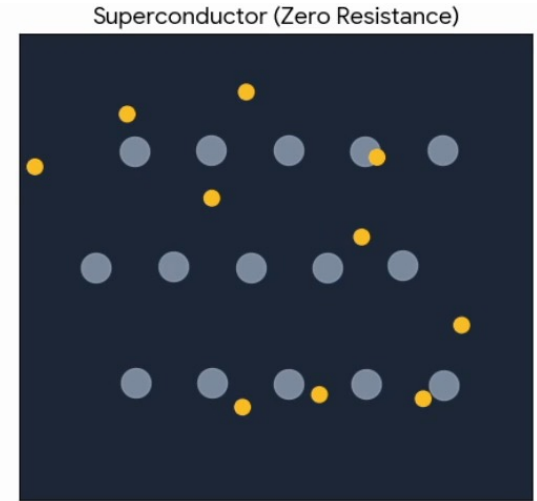
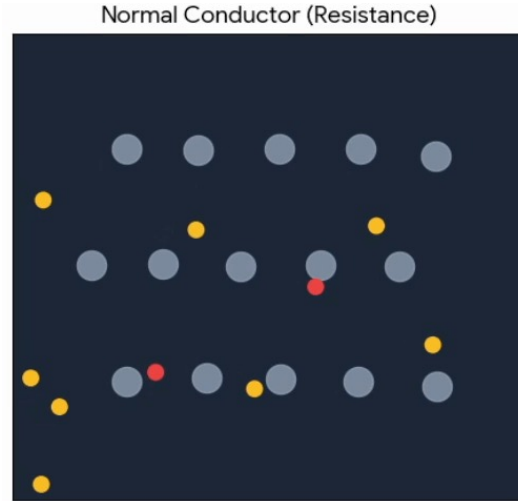
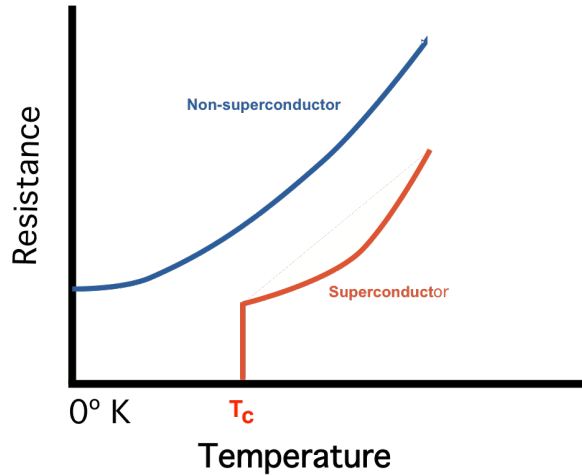
Seitz-Invariant Prediction of Electron Localization Functions from Superposed Atomic Densities via a 3D-Convolutional Network

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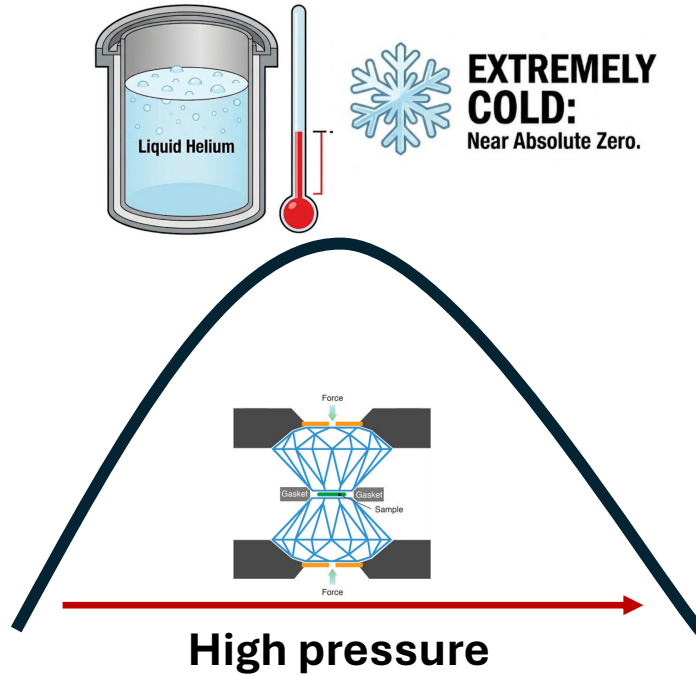
What are superconductors?

Some materials can conduct electricity with **no resistance**!

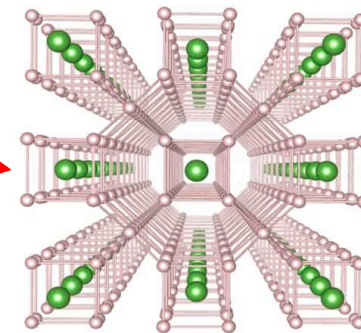
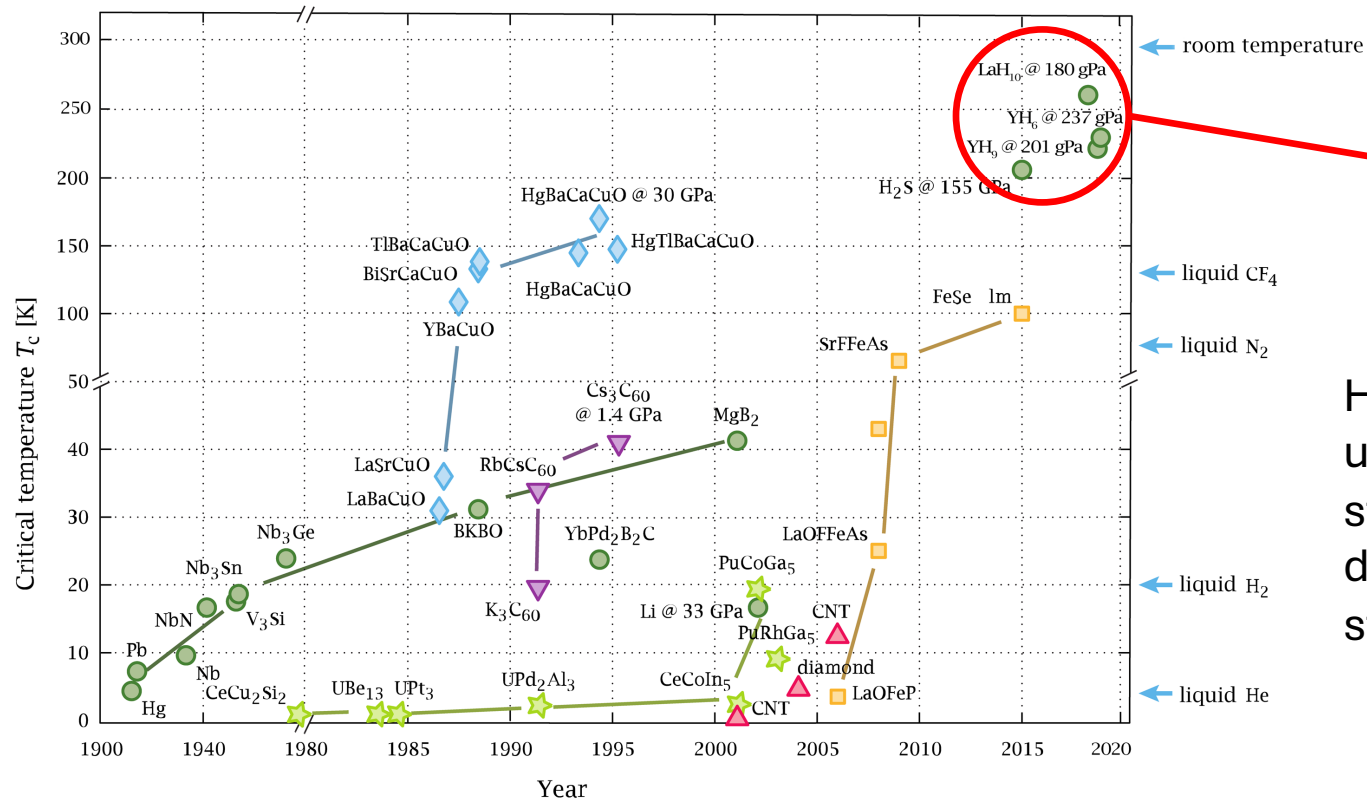


Atom vibrations scatter electrons at high temperature, but help electrons pair and flow with no resistance at low temperature

Superconductors are not science-fiction!



Superhydrides seem promising...



High-pressure chemistry unveils unique material structures leading to discoveries like unusual stoichiometries

The ELF predicts superhydride formation

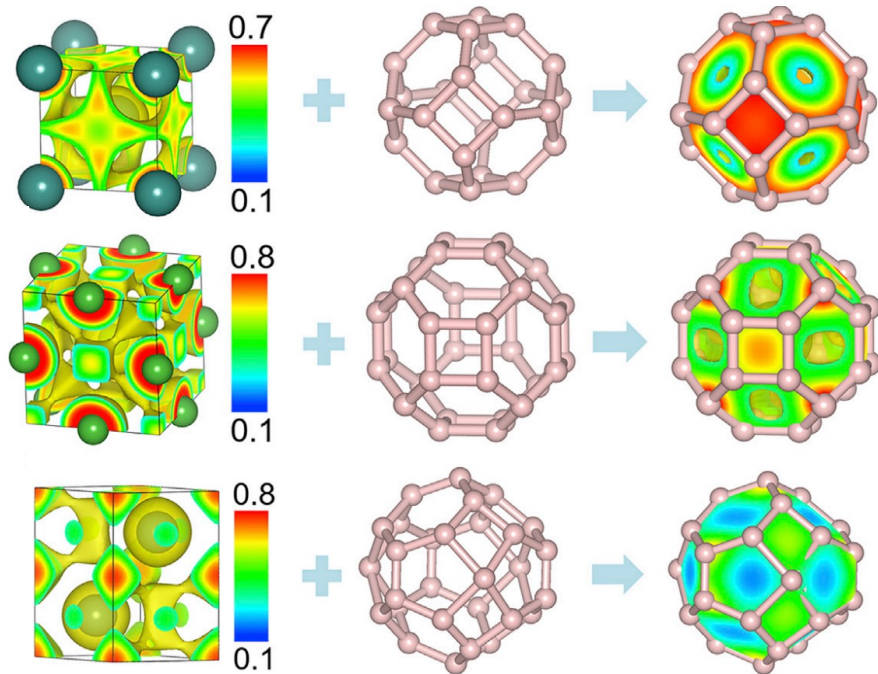
Electron localization function (ELF)
compares local kinetic energy density
to a homogeneous gas

$$\text{ELF}(\mathbf{r}) = \frac{1}{1 + \chi_{\sigma}^2(\mathbf{r})}$$

High value = localized electrons

Low value = delocalization

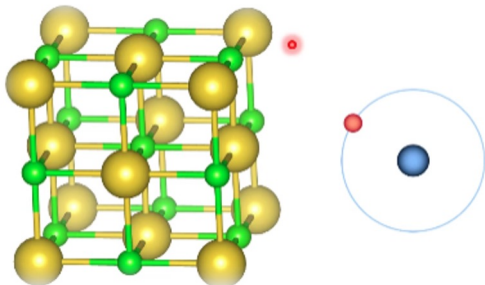
**Hydrogen atoms assemble in
interstitial sites with high localization**



Sun & Miao, *Chem* **2023**, 9, 443–459.

The ELF can be expensive to calculate

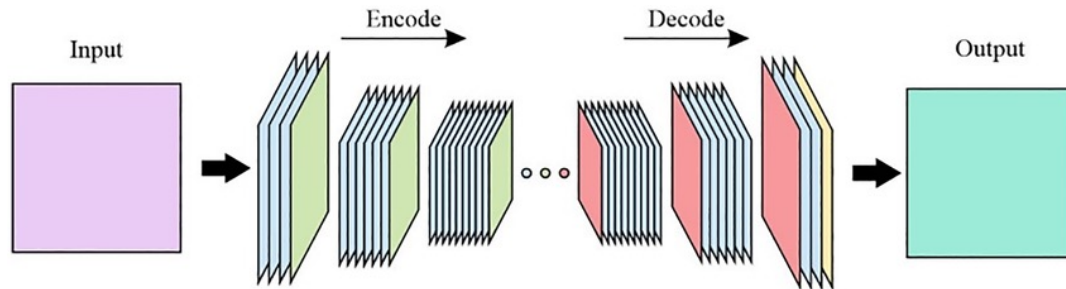
First Principles (QM)



$$-\frac{\hbar^2}{2m}\nabla^2\Psi + V\Psi = E\Psi$$

The time required to run a quantum mechanics-based calculation scales cubically with electron count (N)

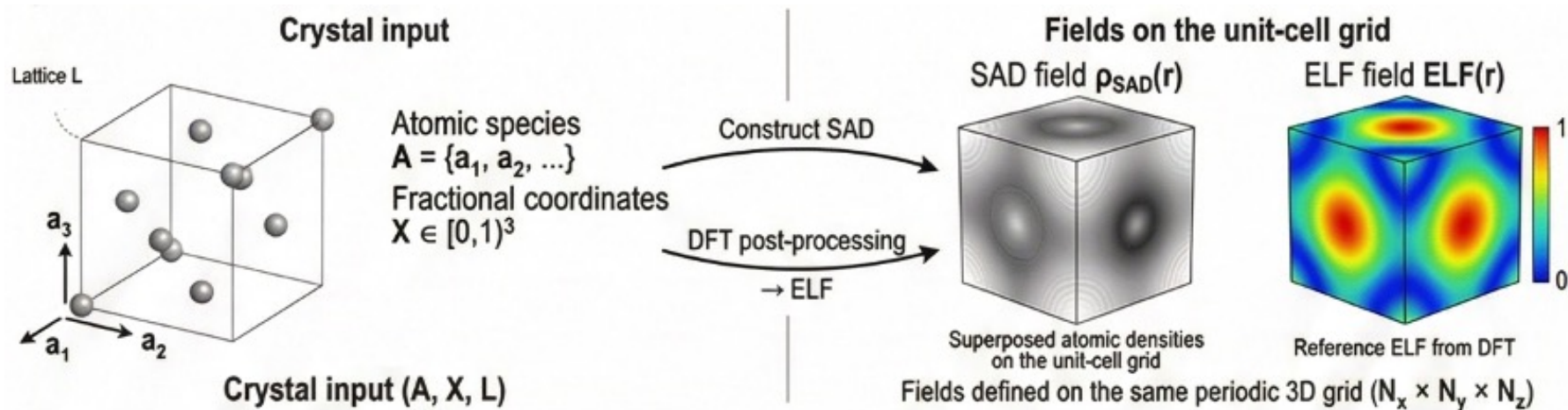
Machine Learning (ML)



Solution: We can **treat the ELF as a 3D image** and train a model to predict it directly

Problem: How do we represent our structure?

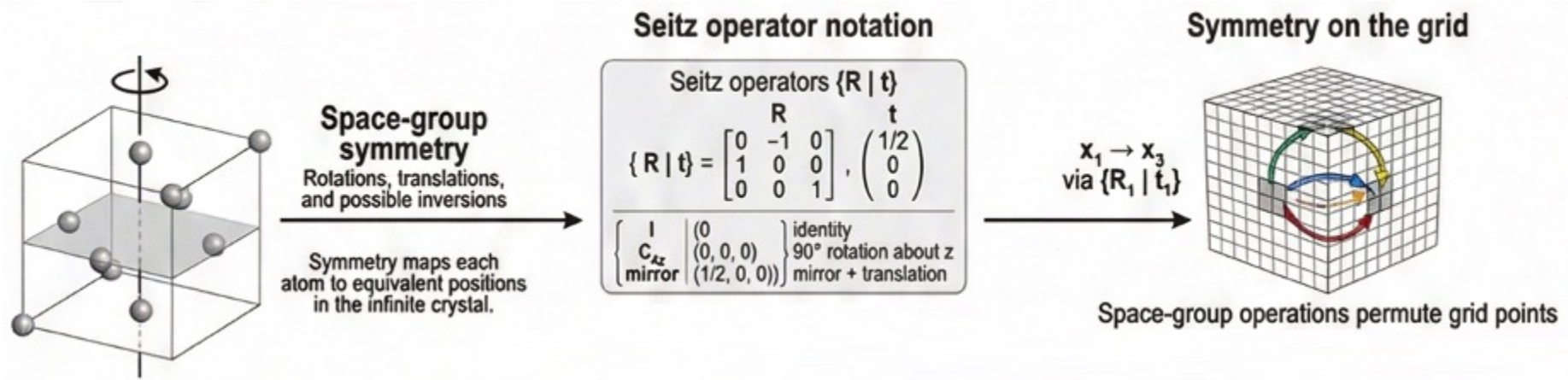
Crystal structure representation



SAD is the electron density obtained by freezing atomic charge density at each atom center without allowing their electron clouds to hybridize or relax into bonded states

Since the SAD and ELF have the same grid size, the **problem is now an image transformation**

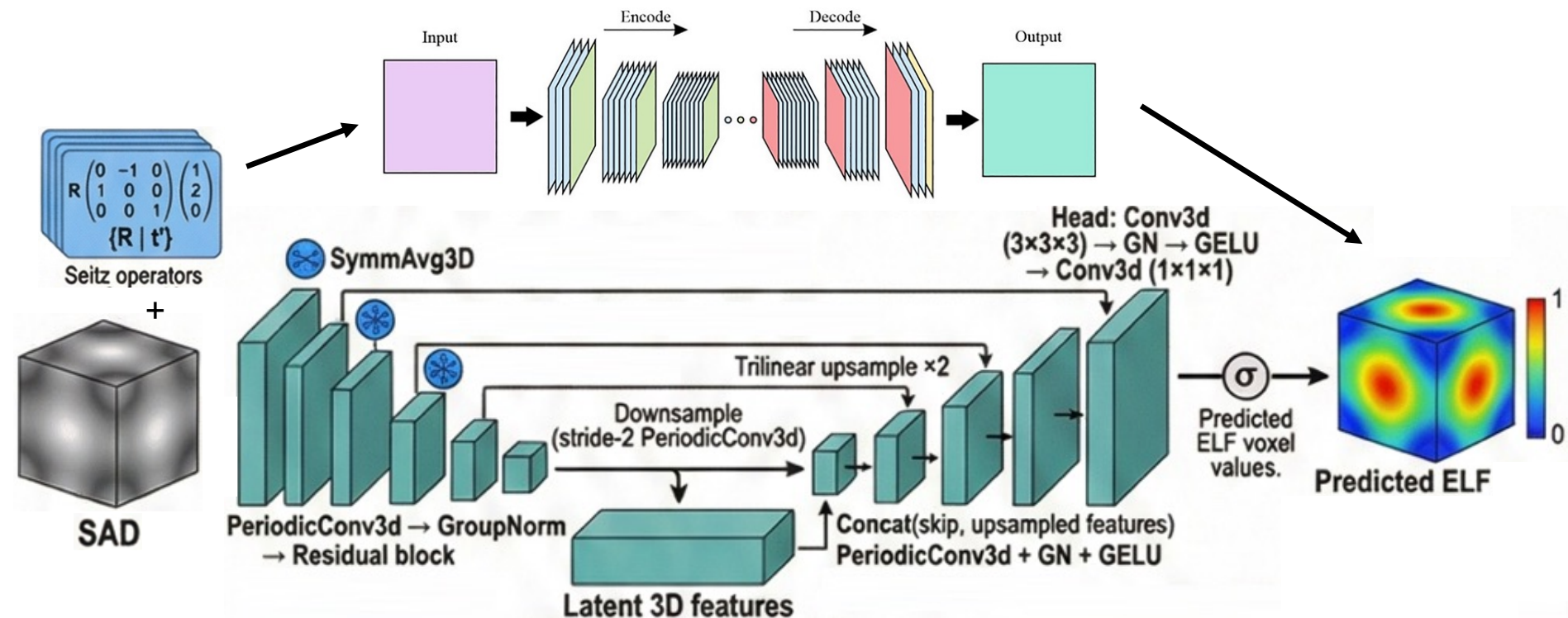
Making the model Seitz-Invariant



Seitz symmetry – Standard notation system for describing the symmetry operations of crystal structures in 3D space groups

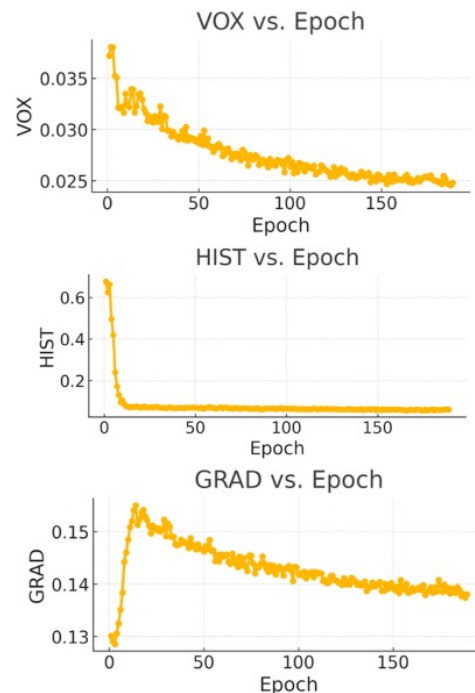
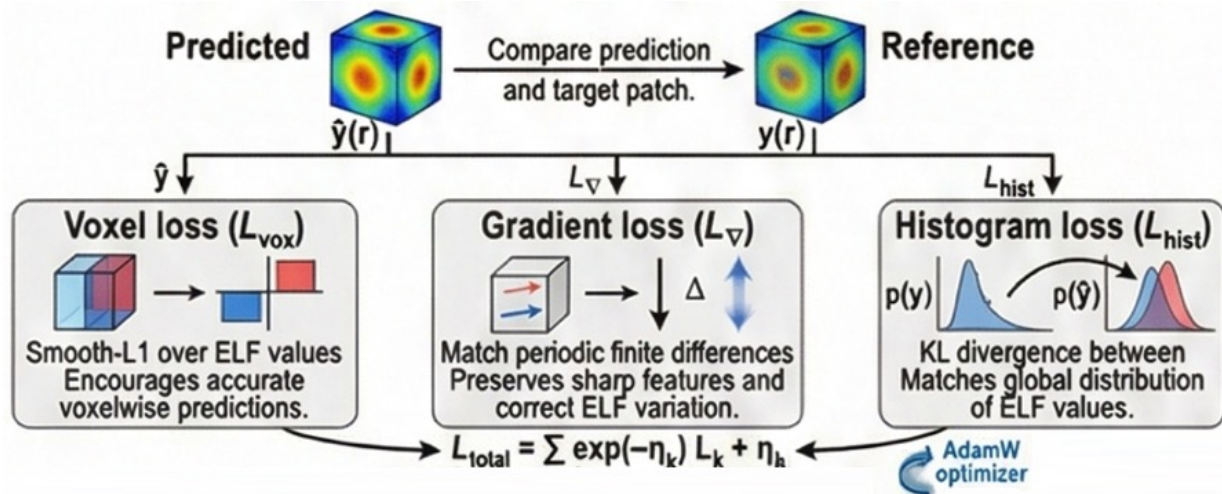
Invariance – Property that guarantees the model makes the same prediction even if the crystal structure is rotated or translated according to its symmetry group

Architecture for a symmetry-aware image transform

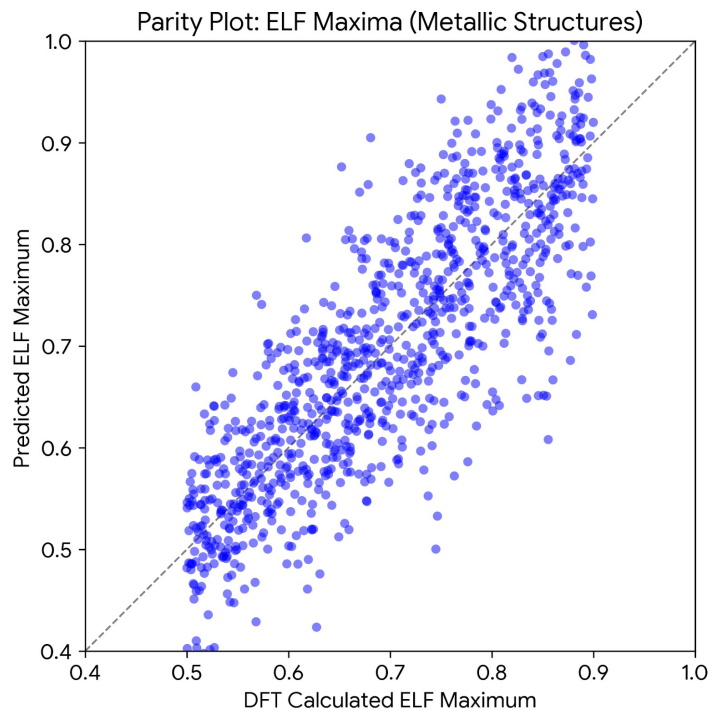
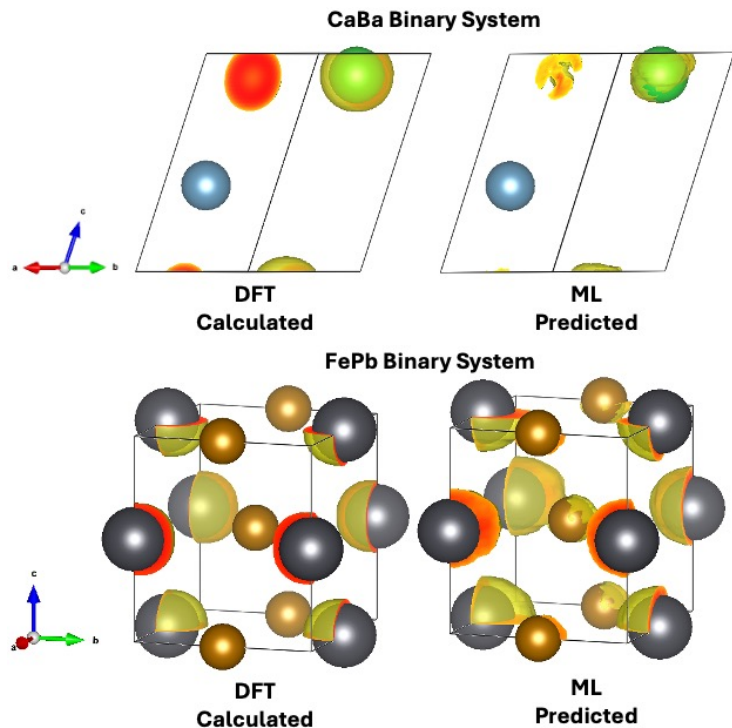


Physics-aware loss function

Kendall Loss: An adaptive weighting scheme that automatically learns how to balance the voxel, gradient, and histogram objectives during training, eliminating the need to manually tune loss weights.



Successful ELF Prediction



Model tested
on subset of
metal lattices

Maxima most
relevant to
prediction of
superhydride
structures

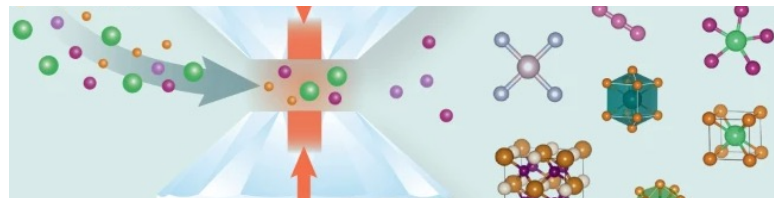
Conclusion

Novel Contributions

- **3D image transformation** bypasses quantum mechanics calculations
- **SAD representations** are used to represent 3D crystals
- **Seitz-Invariance** guarantees predictions respect crystal symmetry
- **Physics-aware optimization** via Kendall loss automatically balances training objectives

Next Steps

- **Extend training set** beyond metals so the SAD→ELF operator supports more chemical space
- **Generate compressed ELF**s to train model on high-pressure regimes
- Use high-pressure model to rapidly evaluate metal lattice ELF's to **discover novel superconductors**



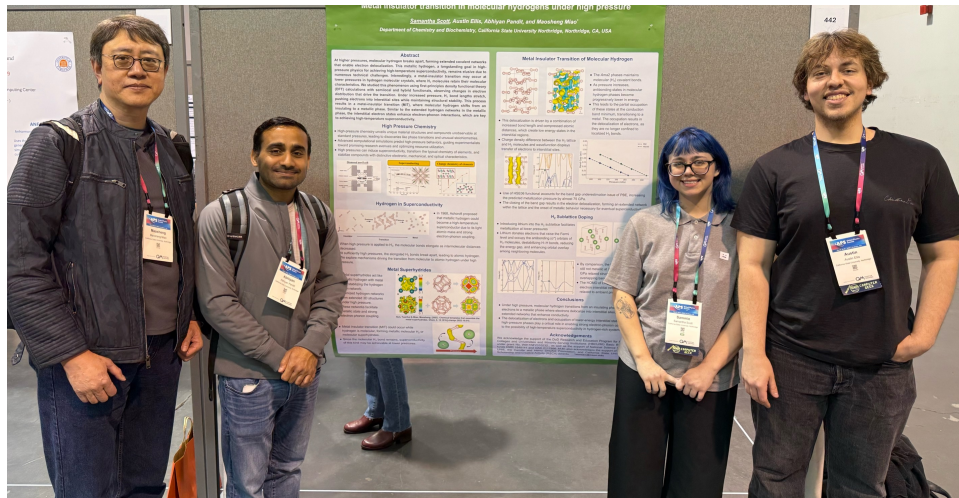
Acknowledgements

Team

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Check out
the paper!

